

# MEGAMINX

12 SIDES · 50 PIECES · ONE SECRET

## SOLUTION GUIDE



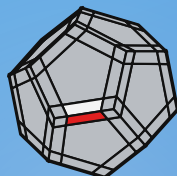
**UNLOCK THE SECRET!**

## DEFINITIONS OF MEGAMINX PIECES

## THE PARTS:

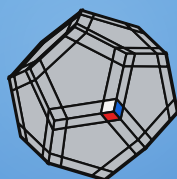
## EDGE PIECES

PIECES WITH TWO (2) COLORS. THERE ARE THIRTY (30) EDGE PIECES. FIVE OF THEM HAVE A WHITE SIDE — THEY MAKE THE WHITE STAR.



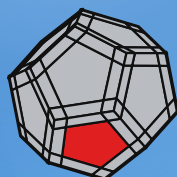
## CORNER PIECES

PIECES WITH THREE (3) COLORS. THERE ARE TWENTY (20) CORNER PIECES.



## CENTER PIECES

PIECES WITH ONE (1) COLOR. THERE ARE TWELVE (12), ONE ON EACH FACE. THEY **DO NOT MOVE** AND REPRESENT THE COLOR OF THEIR FACE.



A megaminx has **12 faces** and **12 colors**. Each color appears on exactly one face. **WHITE** and **GRAY** are on opposite faces — you will start with the white face and finish with the gray one. (Some megaminxes use slightly different colors — use the face colors of YOUR puzzle; the steps are the same.)

## EACH FACE OF THE MEGAMINX IS REPRESENTED BY A LETTER

- U** = **Up Face** — the top face
- F** = **Front Face** — the face facing you, below the top
- R** = **Right Face** — the face just right of the front
- L** = **Left Face** — the face just left of the front
- D** = **Down Face** — the face below and between F and R

A letter with an "i" after it means an **inverted** (counter-clockwise) turn of that face. A letter alone means a clockwise turn (as you look at that face). Every turn is **one click** — one fifth of the way around.

## VERY IMPORTANT

The pictures show exactly which face moves and which way. Hold your megaminx to match the picture before each sequence. Dark gray on a picture means **the color there does not matter**.



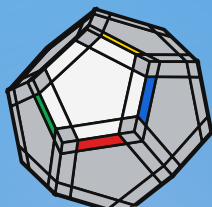
## STAGE 2:

## SOLVE THE WHITE STAR

**HOLDING YOUR MEGAMINX:** Hold your megaminx with the **WHITE** center piece on top (**U**). You will make a white star, and the side color of each star edge must match its side center piece.

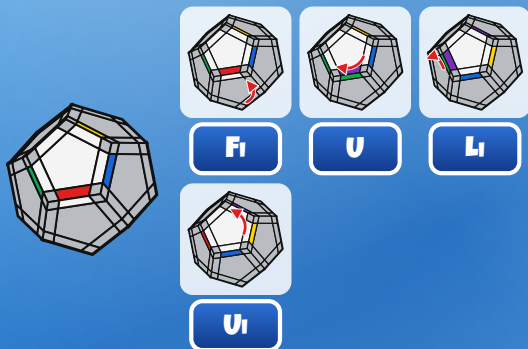
### TIPS:

- Find a white edge in the bottom part of the puzzle. Turning the bottom faces never disturbs finished star edges on top.
- Park it on the **bottom-right edge of the matching side face**, white sticker facing **down-right**; then turn that side face **twice counter-clockwise** — the edge rises into the star.
- Parked on the **bottom-left** with white facing down-left? Turn the side face **twice clockwise** instead. This stage is practice and trial-and-error, just like the white cross on your cube!



**YOUR GOAL**

If an edge is **in the star but Flipped** (white facing out, like the picture), hold the puzzle so that edge is at the front and do this sequence:



**CONGRATULATIONS!**

If the top of your megaminx shows a white star with every side color matching its center, you can move to **Stage 3!**

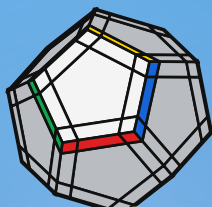
## STAGE 3:

## SOLVE THE WHITE CORNERS

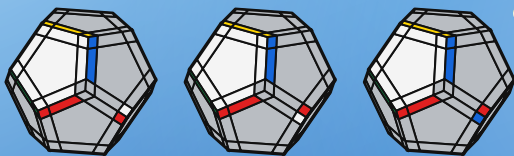
**HOLDING YOUR MEGAMINX:** Keep the **WHITE** star on top (**U**). Now you will fill in the five white corners, one at a time.

### TIPS:

- Find a white corner in the bottom part of the puzzle and turn the bottom faces until it sits **directly below its place** (between the two side colors that match it).
- Hold the puzzle so that corner is at the **front-right**, just below the top, like the three pictures below. The white sticker can face three ways — the sequence works for all of them!



**YOUR GOAL**



Now do this sequence **over and over** (2, 4 or up to 6 times) **UNTIL** the corner pops into place with white on top:



**Ri**



**Di**



**R**



**D**

**NOTE!** If a white corner is stuck in the top layer in the wrong spot or twisted, hold the puzzle so it is at the top-front-right and do the sequence **once** — it will pop out below. Then place it the normal way.

**CONGRATULATIONS!**

If the whole white face and its side colors are solid, you are ready for **Stage 4!**

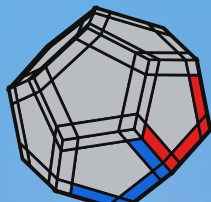
## STAGE 4:

## SOLVE THE SECOND ROW OF EDGES

**HOLDING YOUR MEGAMINX:** Now **tilt the puzzle!** Hold it so the empty edge slot is at the **front** and the face just **below** the slot points **up** — that is your new top (**U**):

### TIPS:

- Turn the new top (**U**) until the edge sits above the front face with its **front color matching** the front center — a straight line, just like on your cube!
- The edge's top color points where it must go: toward (**R**) → sequence 1; toward (**L**) → sequence 2.



### 1) EDGE GOES DOWN-AND-RIGHT:



**U**



**R**



**U<sub>l</sub>**



**R<sub>l</sub>**



**U<sub>l</sub>**



**F<sub>l</sub>**



**U**

**HOLD IT LIKE THIS**



**F**

### 2) EDGE GOES DOWN-AND-LEFT:



**U<sub>l</sub>**



**L<sub>l</sub>**



**U**



**L**



**U**



**F**



**U<sub>l</sub>**



**F<sub>l</sub>**

**CONGRATULATIONS!**

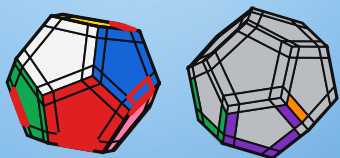
Do all five edges of the second row, then on to **Stage 5!**

## STAGE 5:

## THE MIDDLE BAND

### 1<sup>ST</sup> STEP: PLACE THE FIVE LEANING EDGES

Around the middle of the puzzle there are slanted edge slots. In this step you place **one slanted edge above each petal** — always the one that **leans the same way**, shown outlined in red:



#### TIPS:

- Tilt the puzzle so the empty slanted slot is at the front (second picture) and use exactly the same sequences as Stage 4.
- Go all the way around the puzzle: five slanted edges, one per petal. **Only these five!** The other five slanted slots get filled in the 2<sup>nd</sup> Step together with the corners.



**F**

Then continue straight to the 2<sup>nd</sup> Step on the next page.



## STAGE 5:

## THE MIDDLE BAND

### 2<sup>nd</sup> STEP: CORNER + EDGE PAIRS

**HOLDING YOUR MEGAMINX:** Hold the puzzle so the empty petal pocket (a corner slot plus its last slanted edge slot) is at the **top front-right**. Just like Stage 3 — but the corner brings its edge along for the ride!

#### TIPS:

- Park the corner **below its slot**, and its edge in one of the two parking spots shown — pick by which way the corner's **top color** faces:



repeat sequence 1x



repeat sequence 5x

Then repeat the Stage-3 sequence **until corner and edge BOTH snap in together** (up to 6 times):



R<sub>I</sub>



D<sub>I</sub>



R



D

**NOTE!** Top color facing **down**? Do the sequence **once** and look again — it will match a picture. Piece stuck in a wrong slot? Pop it out as in Stage 3.

## CONGRATULATIONS!

Five pockets done? The middle band is finished — on to Stage 6!



## STAGE 6:

## THE LAST ROW OF CORNERS AND EDGES

### 1<sup>ST</sup> STEP: THE FIVE LOW CORNERS

Tilt the puzzle even further: hold it so an empty low corner slot is at the top front-right (picture). Park the matching corner below it and repeat the Stage-3 sequence until it pops in — **exactly like before**. The staging area is now next to the **GRAY** face, which is still unsolved, so nothing can break.



**R<sub>l</sub>**



**D<sub>l</sub>**



**R**



**D**

### 2<sup>ND</sup> STEP: THE FIVE RIDGE EDGES

Now turn the puzzle so **GRAY** is **on top** — and keep it there until the very end! The last five non-gray edges sit just below the gray face. Use the Stage-4 sequences: turn the gray top until the edge makes its straight line with the front center, then drop it right or left:



**U**



**R**



**U<sub>l</sub>**



**R<sub>l</sub>**



**U<sub>l</sub>**



**F<sub>l</sub>**



**U**



**F**

**CONGRATULATIONS!**

Everything except the gray top layer is done — you are two thirds of the way! On to **Stage 7**.

## STAGE 7:

## MAKE THE GRAY STAR

**HOLDING YOUR MEGAMINX:** **GRAY** stays on top (**U**). Look only at the five **gray edge stickers** on top: how many face up?

### TIPS:

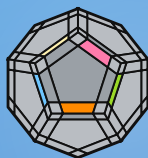
- You will always see exactly **1, 3 or 5** gray stickers facing up. (5 means the star is done — skip to Stage 8!)
- Match your puzzle to one of the pictures, hold it exactly that way, and do the sequence below. Re-count and repeat until the star is complete. The side colors do NOT need to match yet!



1 gray edge: hold it at the **back-right**; sequence once, then re-match.



3 gray; the 2 odd ones **side by side**: hold them at **front + right**; sequence once → star!



3 gray; the 2 odd ones **apart**: hold them at **front + back-right**; sequence **twice** → star!



**F**



**U**



**R**



**U<sub>1</sub>**



**R<sub>1</sub>**



**F<sub>1</sub>**

**CONGRATULATIONS!**

Gray star on top? Excellent — Stage 8!

## STAGE 8:

## PUT THE STAR EDGES IN THEIR PLACES

**HOLDING YOUR MEGAMINX:** **GRAY** on top. Now make every star edge's side color match its center.

### TIPS:

- Spin the gray top (**U**) until **as many edges as possible** match their side centers. At least two will match.
- Hold the puzzle so the two matching edges are at the **front** and **front-left** — the three wrong ones to the right and back, like the picture. Then do the sequence; it swaps the three wrong edges around. You may need it twice.
- If you can't find a hold with two matching edges, do the sequence once from any position and look again.



**CONGRATULATIONS!**

Star perfect all around? **Stage 9** awaits!

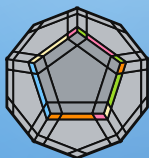
## STAGE 9:

## PUT THE GRAY CORNERS IN THEIR PLACES

**HOLDING YOUR MEGAMINX:** **GRAY** on top. Now move each gray corner **between** the two side colors it matches. Twisted is OK — location is all that counts in this stage!

### TIPS:

- Find corners that are already **between the right colors**. Hold the puzzle so two correct corners are on the **left side** (front-left and back-left) — the three wrong ones at front-right, right and back.
- Do the sequence: it cycles the three wrong corners. Repeat from the same hold if needed.
- Fewer than two correct? Do the sequence once from any position and look again.



**CONGRATULATIONS!**

All five corners between their right colors? One stage to go!

## STAGE 10:

## TWIST THE GRAY CORNERS

**HOLDING YOUR MEGAMINX:** **GRAY** on top. Hold the puzzle with a twisted corner at the **top front-right** and repeat the good old Stage-3 sequence until its gray sticker faces up (2 or 4 times):

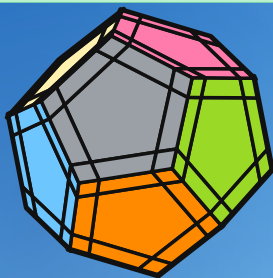


### TIPS:

- Then turn **ONLY the gray top (U)** to bring the next twisted corner to the front-right, and repeat.
- **DON'T PANIC!** The rest of the puzzle will look scrambled while you work — it fixes itself the moment the last corner is twisted right. Trust the moves, just like on your cube!
- When all corners are done, spin the gray top to line everything up.

## CONGRATULATIONS!

**YOU DID IT!** Your megaminx is solved — all 12 colors, all 50 pieces. You have unlocked the secret!



## NOW YOU KNOW THE SECRET...

### FUN FACTS!

The megaminx was invented around 1982, soon after the Rubik's Cube craze began. Early versions were sold under names like the **Hungarian Supernova**.

A megaminx has about **100,669,616,553,523,347,122,516,032,313,645,505,168,688,116,411,019,768,627,200,000,000,000** possible positions — that's a 1 followed by 68 digits. The Rubik's Cube has "only" 43 quintillion!

The world record for solving a megaminx is under **25 seconds**. Every solve in this booklet was checked, move by move, on a computer model of the puzzle — so if you follow the pictures exactly, it always works.

The method you just learned is the same idea as your cube's: solve a star instead of a cross, drop in corners with the Righty moves, pair up the middle, then finish the last layer in four small steps. Bigger puzzle — same secret.



Made for a young puzzler who already unlocked the cube. Happy twisting!